

Nicolas Brosse

ML Researcher — from mathematical foundations to working systems

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Summary

ML researcher working between mathematical research and industrial R&D. PhD in applied mathematics (École Polytechnique) on Langevin sampling and an internship at Google Brain, then six years building deep learning systems in industry — defence, drug discovery, foundation models for biology, LLM systems. Now a postdoctoral researcher on the theory of diffusion models and on human-AI collaboration in mathematical research, through an open framework that I designed and built. I take projects from problem definition to delivery.

Experience

Postdoctoral Researcher · CREST, ENSAE Paris, Palaiseau, France

Nov 2025 – present

Theory of generative models with Arnak Dalalyan; human-AI collaboration in mathematical research.

- Universal local error and realized amplification for the first-order EDM diffusion sampler (preprint, 2026).
- Boundary-layer asymptotics of Gaussian-smoothed singular measures, describing diffusion models in the small-noise regime (preprint, 2026).
- Designed and built an open framework for human-AI collaboration in mathematical research: AI agents search and cross-review proofs, while mathematicians can check every certified claim. First case study: the three 2026 proofs of the KLS conjecture.

Independent AI Research Scientist · Freelance and venture exploration, Paris, France

Jan 2025 – Oct 2025

LLM projects for startups, scoped and delivered end to end.

- **Clarifeye** (Jul–Oct): improved the retrieval and answer quality of a retrieval-augmented generation (RAG) product.
- **Rollstack** (Apr–Jun): built a prototype for evaluating LLM answers with LLM-as-a-judge on an observability platform: test dataset, metrics, integration with the product and engineering teams.
- **Venture exploration** (Jan–Apr): built proofs of concept of RAG-based decision-support tools, with market and technical feasibility studies.

AI Research Scientist · MBZUAI Paris lab, then GenBio AI, Paris, France

Jun 2024 – Dec 2024

Foundation models for biological sequences and structures.

- Led research on tokenization of single-cell RNA and protein data with VQ-VAEs.
- Trained models on 23 TiB of protein structure data on multi-node GPU clusters (SLURM, GCP); set up TM-score evaluation of reconstructed 3-D conformations.

Senior Data Scientist · Iktos, Paris, France

Mar 2022 – Jun 2024

AI for drug discovery.

- Designed and delivered an end-to-end Python package for DNA-encoded library analysis (ingestion, denoising, model training, interactive visualization).
- Trained transformers and graph neural networks for molecular property prediction; studied distance encodings in Uni-Mol pretraining.

AI Research Scientist · Thales, Palaiseau, France

Sep 2019 – Mar 2022

R&D for defence and aerospace systems.

- Object detection, signal processing, anomaly detection and predictive maintenance.
- Uncertainty quantification and interpretability methods to meet reliability requirements of critical systems.

Uncertainty estimation for deep neural network classifiers; parallel training on hundreds of GPUs. Led to the paper on last-layer algorithms (arXiv:2001.08049).

Education

PhD in Applied Mathematics · École Polytechnique	2016 – 2019
<i>Around the Langevin Monte Carlo algorithm: extensions and applications.</i> Advisor: Éric Moulines.	
M.Sc. Probability and Statistics · Université Paris-Saclay	2015 – 2016
Engineering degree (Ingénieur polytechnicien) · École Polytechnique	2012 – 2016

Selected publications

- 2026 N. Brosse, A. S. Dalalyan. Universal local error and realized amplification for the first-order EDM predictor. *arXiv:2610.10190*.
- 2026 N. Brosse, A. S. Dalalyan. Boundary-layer asymptotics for Gaussian-smoothed singular measures. *arXiv:2607.04514*.
- 2020 N. Brosse, C. Riquelme, A. Martin, S. Gelly, É. Moulines. On last-layer algorithms for classification: decoupling representation from uncertainty estimation. *arXiv:2001.08049*.
- 2019 N. Brosse, A. Durmus, É. Moulines, S. Sabanis. The tamed unadjusted Langevin algorithm. *Stochastic Processes and their Applications*.
- 2018 N. Brosse, A. Durmus, É. Moulines. The promises and pitfalls of stochastic gradient Langevin dynamics. *NeurIPS*.
- 2017 N. Brosse, A. Durmus, É. Moulines, M. Pereyra. Sampling from a log-concave distribution with compact support with proximal Langevin Monte Carlo. *COLT*.
- Full list, including EJS 2018 and Comput. Math. Math. Phys. 2024: nbrosse.github.io/research.

Tools for AI-assisted research

Human-AI Mathematics · Open framework for AI-assisted research on a conjecture, and its first case study, the KLS manuscript. human-ai-mathematics.github.io

latexfmt · LaTeX formatter that understands mathematics, published on PyPI.

Service

X-AI Board member of the Polytechnique alumni AI group (1,000+ members), 2023–2024; organized four events on AI in health and biology.

Skills

Research	Generative models (diffusion, flow matching), Langevin sampling, functional inequalities, uncertainty quantification, evaluation of ML and LLM systems
Engineering	Python, PyTorch, multi-GPU and multi-node training (SLURM, GCP), transformers, GNNs, VQ-VAEs, RAG, multi-agent workflows (Claude Code, Codex), LLM-as-a-judge evaluation
Languages	French (native), English (professional working proficiency)